ASIA PACIFIC UNIVERSITY OF TECHNOLOGY AND INNOVATION
RESPONSIVE WEB DESIGN & DEVELOPMENT
GROUP ASSIGNMENT

Intake :
Lecturer :
Date Due : Week 12
Date Assigned : Week 2

Course Learning Outcomes	Assignment Component
CLO1 Explain the Internet Infrastructure and web page development (A3, PL08)	Documentation
CLO2 Develop a responsive website using client-side scripting applying HCI principles for various user clients (C3, PLO3)	Implementation

ASSIGNMENT BRIEF

Your Web Development Team is tasked with the development of a fully functional web application. The application's purpose is to provide online assessment activities such as Interactive Quizzes, Skill Assessments, Knowledge Tests, etc., aimed at engaging and evaluating the expertise of targeted users (i.e., students) in various subjects.

You have the freedom to choose the title of the application. However, it should adhere to the following guidelines:

- **Client-side scripting:** The application should effectively utilize client-side scripting languages, specifically HTML, CSS, and JavaScript.
- **Server-side scripting:** PHP should be used as the server-side scripting language.
- **Database Management:** MySQL should be employed for database management.
- **Interface Design:** The interface design should adhere to good design principles.
- **Responsive Website:** Website should be responsive where it should display properly in smaller screen and bigger screen like mobile, tablet and desktop.
- **Navigation:** The application should feature a fully functional and user-friendly navigation system.

- **Multimedia Content:** Where necessary, appropriate multimedia content should be incorporated.

The web application should demonstrate database processing capabilities, including user login functionality and the ability to insert, read, update, and delete records. Your web application must be designed to accommodate various types of registered users. At a minimum, there should be two distinct user roles within the application. Form validation should be implemented to ensure data integrity.

Please note that your team has complete discretion in determining the scope and contents of the chosen title. This discretion should be reflected in the application's design and functionality. Please select your group number on the Moodle course page. Each group should consist of **3 to 4 members**.

REQUIREMENTS

The usage of Web page creation tools and templates are NOT permitted. You have the option of developing the website using the following tools:

- Visual studio code
- Notepad++
- Sublime

DOCUMENTATION (50%)

1. **Proposal (Due Week 5)**
    - Title of the Web site.
    - Objectives of the project.
    - Process Flow chart.
    - ERD Diagrams.
    - Wireframe.
    - Navigational Structure.

2. **Report (Due Week 13)**
   Write a report (2000 words) which includes the following:

- Format:
    - Typeface: Please use Times New Roman. You may use bold, italic, and underlined text for emphasis and to enhance readability.
    - Font size: The standard font size should be 12, except for titles and headings.
    - Spacing: Maintain a spacing of 1.5 lines between the texts within a paragraph.
    - Alignment: The text should be justified.
    - Headers and footers can be utilized for additional information.
    - The document should be numbered for easy reference.
    - A cover page is required. Your cover page should include the following details:
        - Your Name and ID
        - Intake code
        - Subject
        - Project Title
    - Recommended Content
        - **Table of Contents:** The table **of** contents should list each topic title along with its

corresponding page number.

- **Introduction**/Project **Plan:** This section should introduce your project, including its objectives, scope, end-user specifications, and the major functions of the web application.

- **Background Analysis & Requirement Gathering** includes an understanding of the overall context in which the website will operate. The analysis may include studying existing websites or applications to identify best practices, industry trends, and potential challenges. Requirement Gathering focuses on capturing and documenting the specific requirements and expectations of the client. It involves engaging with stakeholders, such as clients, end-users, and subject matter experts, to elicit their needs and preferences.

- **Design:** Create an ERD model, wireframes, a system flowchart, a navigational structure, and a site focus that outlines your objectives for the site based on the principles presented in the lecture. Additionally, you must explain the design choices you made and the rationale behind them.

- **Implementation** - Discuss the steps on how you implement the system.

- **Conclusions –** This section reviews and evaluates the works you have conducted. You can validate the website whether it has achieved the formulated objectives. Also, you are required to discuss limitations of the website and ways to address these issues in the future.

- **References:** A referenced report, including bibliography, on the future of the internet, mainly focusing on web sites and their use to business and future technology that are being developed for business use. Strictly APA style referencing

IMPLEMENTATION (50%) (Due Week 13)

The website should include the following:

- Appropriate graphics, including some you have created.
- All website file should be in HTML 5 format and responsive depend on the devices size for example mobile and desktop.
- Style sheets can be embedded, inline or external.
- Valid and appropriate usability / accessibility for the target audience
- Website must be responsive and display correctly in smaller screen and bigger screen like mobile, tablet and desktop.

This section will be assessed by a **15 minutes** demonstration.

SUBMISSION

1. Documentation must be submitted to Moodle.

2. All other necessary files to run the website and project documentation must be submitted to Moodle as a zipped folder.

PRESENTATION

You are required to present your website to the lecturer, and the presentation will be judged accordingly. You can use any of the presentation software, e.g. PowerPoint.

ASSESSMENT CRITERIA

The assignment is weight at 100%. Group efforts will be assessed based on the following criteria.

No	Criteria	Mark Allocated	Score	Grade
	Documentation (50%)			
1	Introduction **Criteria:** • **Clarity of Introduction:** Evaluate the clarity and conciseness of the project introduction. **(4)** • **Purpose Statement:** Assess if the introduction effectively outlines the purpose and scope of the project.**(4)**	10	0	
2	Background Analysis & Requirement Gathering **Criteria:** • **Depth of Analysis:** Evaluate the depth and thoroughness of the background analysis related to the project. **(4)** • **Requirement Gathering:** Assess how well project requirements are documented and justified. **(4)**	10	0	0
3	Storyboard and Modeling **Criteria:** • **Storyboard Quality**: Evaluate the clarity and effectiveness of the storyboard in illustrating the user experience flow. **(4)** • **Modeling**: Assess the accuracy and completeness of any models (e.g., data models, process models) included in the documentation. **(4)**	10	0	F-
4	Screenshots, Evaluation & Future Enhancements **Criteria:** • **Screenshots:** Evaluate the relevance and clarity of screenshots included to illustrate key aspects of the project.**(2)** • **Evaluation:** Assess the effectiveness of the project evaluation section in analyzing strengths and weaknesses.**(4)** • **Future Enhancements:** Evaluate the feasibility and creativity of suggested future enhancements.**(2)**	10	0	

5	Conclusion, References & Documentation Standard **Criteria:** • Conclusion: Evaluate the completeness and effectiveness of the project conclusion. **(4)** • References: Assess the quality and relevance of the references cited throughout the documentation**.(2)** • Documentation Standard: Evaluate adherence to documentation standards (e.g., formatting, citation style).**(2)**	10	0	
	Implementation (60%)			
6	Responsive web page presentation (Layout and Design), Navigation Structure & Usability **Criteria:** • **Layout and Design:** Evaluate the visual appeal, consistency, and aesthetics of the website layout. **(4)** • **Navigation Structure:** Assess the ease of navigation, clarity of menus, and accessibility of links.**(3)** • **Usability:** Evaluate how user-friendly the interface is, considering intuitive design principles and user expectations.**(3)**	10	0	
7	Multimedia Elements & Originality **Criteria:** • **Multimedia Integration:** Evaluate the use of multimedia elements such as images, videos, and audio.**(5)** • **Originality:** Assess the creativity and uniqueness of the multimedia content used.**(5)**	8	0	0
8	HTML & CSS **Criteria:** • **HTML:** Evaluate the use of HTML markup for structure, semantics, and accessibility.**(5)** • **CSS:** Assess the use of CSS for styling, layout management, responsiveness, and adherence to best practices.**(5)**	10	0	F-
9	Features using JavaScript **Criteria:** • **JavaScript Functionality:** Evaluate the implementation of interactive features and functionalities using JavaScript.**(5)** • **Performance:** Assess the efficiency and responsiveness of JavaScript code, considering loading times and user interactions.**(5)**	8	0	
10	Features using PHP **Criteria:**	8	0	

	• **Functionality:** Evaluation the implementation of the PHP whether the PHP scripts do what it intend to do.**(6)** • **Code Quality:** Evaluate the structure and readability of the code.**(2)** • **Security:** Assess the data validation and error handling**(2)**			
11	Presentation	6	0	F-
	Total Mark:	**100**	**0**	**F, Fail**
	Percentage (%):	**100**	**0**	

GRADING CRITERIA

DISTINCTION: 75% - 100%

Distinctive work will exhibit a high level of penetration and discernment in analysis and outstanding cogency and clarity in communication, design, implementation and testing of the prototype. The supporting report should be exceptionally well-substantiated in its analysis, reasoning, and conclusions within each area.

In addition, at the advanced level the student will demonstrate an exceptional ability to synthesize innovative perspectives of the topic with work from other parts of the discipline.

CREDIT: 65% - 74%

A student should be able to show further evidence of knowledge, understanding and application of the module material, through analysis, design, implementation and testing of the prototype. The supporting report should be well written and supported throughout in its analysis, reasoning and conclusions.

An advanced level will be indicated by the student's reference to and integration of material from previous programmes of study.

PASS: 50% – 65%

The student will show the ability to select and apply appropriate techniques for analysis, design, implementation and testing of the prototype.

There should be satisfactory evidence of knowledge, understanding and application of the module material in the supporting report. Generally, the report will be less effective in conveying meaning

MARGINAL FAIL: 40%-49%

There should be basic evidence of knowledge, understanding and application of the module domain shown in the application of the design, implementation and testing of the prototype. Generally, the report will be less effective in conveying meaning.

FAIL: 0%-39%

Work at this level will generally be of low standard where it may even fail to meet at least 50% of the basic requirements listed above. Work that is not documented adequately, lacks referencing and does not display proper program structure would fall in this category. During demonstration, the student is not able to explain briefly the work that has been done.